LPMF — Presentation Script

Load Profile Management and Forecasting

Slide-by-slide speaker notes in Hinglish, with formulas, worked examples and viva Q&A

Team

Vansh Sachan

Animish Gupta

Ayush Rathore

Lavish Rawat

Under the guidance of

Dr. Ujjal Sur (Supervisor)

Dr. Arjun Tyagi (Co-Supervisor)

Department of Electrical Engineering · Faculty of Technology · University of Delhi

Poori kahani 1 minute mein

Agar examiner kahe 'ek minute mein batao project kya hai', to yeh bolo (~60 second):

> Delhi ki bijli ki demand har saal garam, peaky aur mehngi hoti ja rahi hai. 29 June 2026 ko Delhi ka peak 8,748 MW tak gaya. DISCOM ko har din agle din ki demand 96 blocks mein, yaani har 15 minute ke liye, batani padti hai, aur galti hone par CERC ke DSM rules ke under paisa lagta hai. Humne ek dashboard banaya jo sirf forecast nahi dikhata, balki uska nateeja bhi dikhata hai. Real Delhi SLDC load data (April 2023 se January 2026) aur Open-Meteo weather ko saaf kiya, 15-minute blocks banaye, aur ek LightGBM model train kiya jo agle din ke 96 blocks predict karta hai. Test period par iski error (MAPE) 3.93% hai, jabki simple seasonal-naive baseline ki 5.16%. Har forecast ke saath ek P10–P90 band bhi hai, jisme 84.4% actual values aayi, target 80% tha. Dashboard ke 12 pages mein se 6 live hain: Overview, Load Analytics, Weather Impact, Forecast, DSM ₹ Calculator aur Transformer Health. Baaki 6 future scope hain. Har number par label hai ki woh REAL hai, SYNTHETIC hai ya ILLUSTRATIVE. Aur hum apni limitations bhi khul ke batate hain: Diwali jaise din par error zyada hai, aur test set mein garmi ka peak season nahi hai.

Timing table

Slide	Title	Time (sec)	Clock (end)
1	Title	35	0:35
2	Problem Statement	75	1:50
3	Gap and USP	70	3:00
4	Dashboard: 6 Live, 6 Future Scope	60	4:00
5	Dataset & Variables	60	5:00
6	Model Accuracy & Fit Check	75	6:15
7	Overview	60	7:15
8	Load Analytics	65	8:20
9	Weather Impact	65	9:25
10	Forecast	70	10:35
11	DSM ₹ Calculator	60	11:35
12	Transformer Health	60	12:35
13	References	25	13:00
14	Thank You	25	13:25
	Total	**805**	**13:25 (≈ 13.4 min)**

Speed ka andaaza: ~140 shabd per minute. Viva ke liye 3–5 minute alag rakho. Agar time kam ho, slide 5 (Dataset), slide 6 (Accuracy) aur slide 10 (Forecast) poori bolo; baaki ki script chhoti kar sakte ho.

Slide 1 — Title



Slide par kya hai

Project ka naam (Load Profile Management & Forecasting), Faculty of Technology, University of Delhi, team ke chaar naam aur dono guides: Dr. Ujjal Sur (Supervisor) aur Dr. Arjun Tyagi (Co-Supervisor).

Bolne wali script (~35 sec)

> Good morning sir/ma'am. Hum aaj apna project present kar rahe hain: Load Profile Management and Forecasting, yaani Delhi grid ke liye ek AI-integrated load intelligence dashboard. Main hoon Vansh Sachan, aur mere saath hain Animish Gupta, Ayush Rathore aur Lavish Rawat. Yeh project humne Dr. Ujjal Sur aur Dr. Arjun Tyagi ki guidance mein kiya hai. Agle kuch minute mein hum batayenge ki problem kya hai, humne kya banaya, aur har number kahan se aaya.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Project aur team ka introduction.

Kyu kiya: Audience ko pehle hi pata chal jaye ki topic power system aur machine learning dono ka hai.

Kaise kiya: Ek line ka subtitle: 'AI-integrated load intelligence dashboard for the Delhi grid'.

Viva mein pooche ja sakte hain

Q: Project ka one-line summary kya hai?

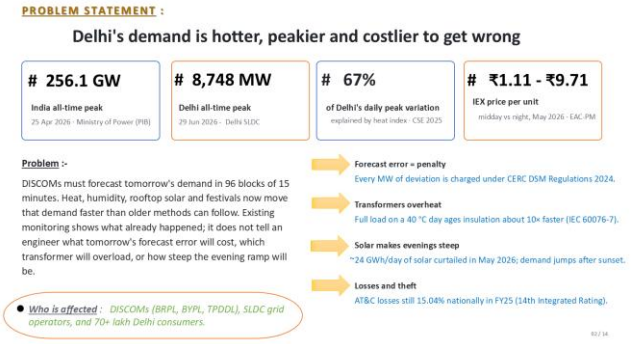
A: Real Delhi data se agle din ki 96-block load forecast, aur us forecast ko peak alert, DSM ₹ aur transformer ageing jaise engineering nateejon tak le jaana.

Q: Isme AI kahan hai?

A: Forecast LightGBM model se aata hai, jo ek machine learning model hai. Baaki pages engineering formulas hain.

Agli slide par: *To chaliye, pehle dekhte hain ki Delhi ki demand ki problem itni serious kyun hai.*

Slide 2 — Problem Statement



Slide par kya hai

Chaar bade numbers: India ka all-time peak 256.1 GW, Delhi ka 8,748 MW, heat index se 67% daily-peak variation (CSE 2025), aur IEX price ₹1.11 se ₹9.71 per unit. Neeche problem aur chaar asar: penalty, transformer heating, steep evening, losses.

Bolne wali script (~75 sec)

> Delhi ki demand garam, peaky aur galat hone par mehngi hai. 25 April 2026 ko India ka peak 256.1 GW tha, aur 29 June 2026 ko Delhi ka record 8,748 MW. CSE ki 2025 study kehti hai ki Delhi ke daily peak ke variation ka lagbhag 67% heat index se samjha ja sakta hai. Heat index matlab temperature aur humidity ko milake 'kitni garmi mehsoos hoti hai'. DISCOMs ko agle din ki demand 96 blocks of 15 minutes mein batani padti hai. Agar forecast galat ho, to CERC DSM Regulations 2024 ke under har MW deviation ka charge lagta hai. Saath hi, 40 degree ke din full load par transformer ki insulation lagbhag 10 guna tez umar kha jaati hai, solar ki wajah se shaam ki demand tezi se chadhti hai, aur AT&C losses abhi bhi 15.04% hain. Aaj ke monitoring tools sirf bata dete hain ki kya ho chuka; yeh nahi batate ki kal ki galti ka kya nateeja hoga.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Problem ko numbers ke saath define kiya.

Kyu kiya: Sirf 'forecast karna hai' kehne se importance samajh nahi aati; paisa, equipment aur grid par asar dikhana zaroori tha.

Kaise kiya: Har number ke saath official source likha: PIB, Delhi SLDC, CSE, EAC-PM, IEC 60076-7, 14th Integrated Rating.

Formula / number ka matlab

96 blocks ka matlab

1 din = 24 ghante × 4 blocks per ghanta = 96 blocks, har block 15 minute ka. DSM settlement isi block level par hota hai, isliye forecast bhi 96 blocks ka chahiye.

Viva mein pooche ja sakte hain

Q: 67% wala number aapka hai?

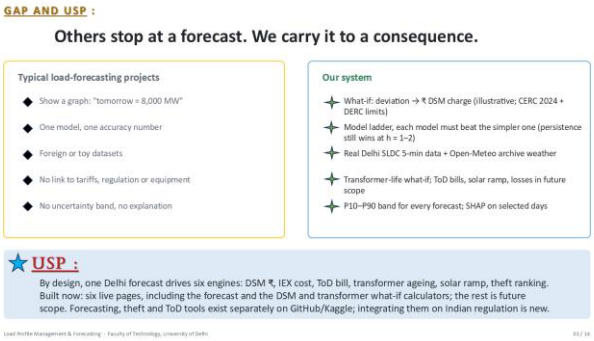
A: Nahi, yeh CSE 2025 ka hai aur daily peak ke liye hai. Hamara apna fit har 15-min block par $r^2 = 0.61$ deta hai. Dono alag metric hain, isliye directly compare nahi karte.

Q: 8,748 MW aapke data mein hai?

A: Nahi. Hamara data 12 January 2026 par khatam hota hai, aur yeh peak 29 June 2026 ka hai. Data ka maximum 8,581.6 MW (19 June 2024) hai. Isliye 8,748 MW sirf alert limit ki tarah use kiya.

Agli slide par: *Ab sawal yeh hai ki baaki projects yeh problem kyun solve nahi karte; yeh agli slide mein.*

Slide 3 — Gap and USP



Slide par kya hai

Do column: typical load-forecasting projects kya karte hain, aur hamara system kya alag karta hai. Neeche USP box.

Bolne wali script (~70 sec)

> Zyadatar forecasting projects ek graph dikha ke ruk jaate hain, jaise 'kal 8,000 MW'. Ek model, ek accuracy number, aur aksar foreign ya toy dataset. Hum forecast ko uske nateeje tak le jaate hain. Pehla, model ladder: har complex model ko apne se simple model ko beat karna padta hai, warna hum use nahi rakhte. Aur hum imaandari se batate hain ki h = 1–2, yaani 15–30 minute aage, simple persistence abhi bhi jeet jaata hai. Doosra, data real hai: Delhi SLDC ka 5-minute data aur Open-Meteo weather. Teesra, har forecast ke saath P10–P90 band hai, aur selected din par SHAP explanation. Chautha, DSM ₹ aur transformer life ke what-if calculators. Hamara USP: design ke hisaab se ek Delhi forecast chhe engines chalata hai. Abhi chhe live pages bane hain; ToD bill, solar ramp aur losses future scope hain. Alag-alag tools GitHub aur Kaggle par milte hain, lekin unhe Indian regulation par ek saath jodna naya hai.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Existing kaam aur hamare kaam ka farak saaf kiya.

Kyu kiya: Examiner ka pehla sawal hota hai 'isme naya kya hai?'.

Kaise kiya: Paanch point side-by-side compare kiye, aur jo bana nahi hai use 'future scope' likha.

Viva mein pooche ja sakte hain

Q: Persistence kya hai?

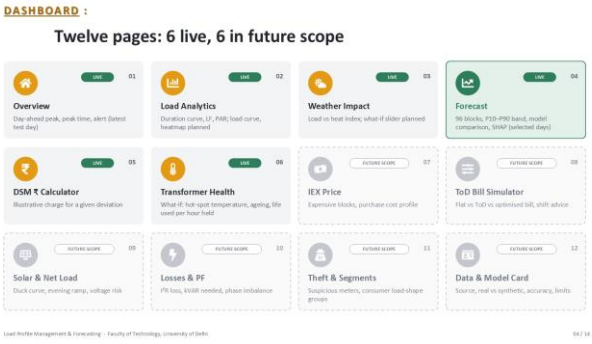
A: Sabse simple forecast: agla block = abhi wala block. 15 minute mein load bahut kam badalta hai, isliye bahut chhote horizon par yeh bahut accurate hota hai.

Q: SHAP kya hai?

A: Ek method jo batata hai ki kis feature ne forecast ko kitna upar ya neeche dhakela. Jaise 'kal ka load' sabse bada driver nikla.

Agli slide par: *Ab dekhte hain ki yeh sab dashboard par kaise dikhta hai.*

Slide 4 — Dashboard: 6 Live, 6 Future Scope



Slide par kya hai

12 cards. Pehle 6 (Overview, Load Analytics, Weather Impact, Forecast, DSM ₹ Calculator, Transformer Health) rangeen aur 'LIVE'. Baaki 6 (IEX Price, ToD Bill Simulator, Solar & Net Load, Losses & PF, Theft & Segments, Data & Model Card) grey aur 'FUTURE SCOPE'.

Bolne wali script (~60 sec)

> Dashboard mein 12 pages hain, aur har page ek operational sawal ka jawab deta hai. Chhe pages live hain aur real API se chalte hain. Overview batata hai kal ka peak kitna aur kab hoga, aur alert hai ya nahi. Load Analytics batata hai demand kitni peaky hai. Weather Impact batata hai garmi se kitna load badhta hai. Forecast mein 96 blocks, uncertainty band aur model comparison hai. DSM ₹ Calculator aur Transformer Health what-if calculators hain. Baaki chhe pages, jaise IEX Price, ToD Bill Simulator, Solar & Net Load, Losses & PF, Theft & Segments aur Data & Model Card, future scope hain. Humne unhe jaan-boojh kar grey dikhaya hai taaki koi confuse na ho ki kya bana hai aur kya nahi.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Poore system ka map ek slide par.

Kyu kiya: Live aur planned kaam ko alag dikhana imaandaari hai; examiner ko sirf bani hui cheez par judge karna chahiye.

Kaise kiya: Next.js frontend, FastAPI backend; live cards rangeen, future cards grey dashed border ke saath.

Viva mein pooche ja sakte hain

Q: Data & Model Card future scope mein kyun hai?

A: App mein uska route hai, lekin hum use poora bana hua page nahi maante, isliye future scope mein rakha.

Q: Live pages ka data kahan se aata hai?

A: FastAPI se. Har response mein provenance (REAL / ILLUSTRATIVE) aur source string hoti hai.

Agli slide par: *Har page ka number kahan se aata hai, yeh agli slide batati hai: dataset.*

Slide 5 — Dataset & Variables



DATASET & VARIABLES :

Where each number comes from, and what goes in

Feature	Source	Key variables	Real / synthetic	How (cleaning, or why synthetic)
Overview	Delhi SLDC 15-min load (Kaggle, Apr 2023 – Jan 2026), cleaned; day-ahead LightGBM forecast (test period)	predicted_mw ×96 blocks, limit_mw = peak_mw, peak_time, peak_block, alert	REAL	Causal Hampel outlier filter, tiered gap-filling, chronological train/val/test split
Load Analytics	Delhi SLDC 15-min load (Kaggle, Apr 2023 – Jan 2026), cleaned	load_mw, ts_ist + load_factor, par, p_avg_mw, p_max_mw, load_duration_curve	REAL	Same cleaning: missing blocks skipped, never counted as zero load
Weather Impact	Delhi SLDC load, cleaned + Open-Meteo weather (lat 28.6139, lon 77.2090)	load_mw, heat_index_c (from temp_c, rh_pct), ts_ist + beta_mw_per_c, r_squared, n_blocks	REAL	Heat index from temp + humidity (Rothfusz); OLS refit on all 96,708 non-missing blocks
Forecast	Delhi SLDC load, cleaned + Open-Meteo weather; LightGBM backtest	ts_target, block, predicted_mw, p10_mw, p90_mw, actual_mw; features tlag_1day, night_temp_c	REAL	Chronological split (test 13 Jul 2025 – 12 Jan 2026); split-conformal P10–P90 band
DSM ₹ Calculator	User what-if input (sliders)	deviation_mw, schedule_limit_mw, normal_rate_per_mwh, excess_multiplier + penalty_rupees	ILLUSTRATIVE	Not data-driven: formula applied to user-supplied values, always tagged Illustrative
Transformer Health	User what-if input (sliders)	hotspot_temp_c, hours = ageing_rate_v, life_hours_used	ILLUSTRATIVE	Not data-driven: formula applied to user-supplied values, always tagged Illustrative

Future-scope modules only (Theft & Segments, Solar & Net Load): synthetic category profiles (residential, commercial, industrial, agricultural, EV) from a parametric model: base + daily pattern + weekly pattern + heat effect + noise + event bumps, validated against the real aggregate shape (peak timing, load-duration curve, autocorrelation). The six live features above run entirely on real data, except the two Illustrative calculators.

Load Profile Management & Forecasting – Faculty of Technology, University of Delhi 05 / 14

Slide par kya hai

Table: chhe live features, har ek ka source, key variables (exact field names), REAL ya ILLUSTRATIVE tag, aur cleaning ya synthetic hone ka kaaran. Neeche synthetic data ka note.

Bolne wali script (~60 sec)

Yeh slide batati hai ki har number kahan se aata hai. Pehle chaar features, yaani Overview, Load Analytics, Weather Impact aur Forecast, poori tarah REAL data par chalte hain: Delhi SLDC ka load, April 2023 se January 2026, aur Open-Meteo weather Delhi ke coordinates par. Variables wahi hain jo code mein hain, jaise load_mw, heat_index_c, temp_c aur rh_pct. Cleaning causal hai, yaani har decision sirf past data dekh kar hota hai: Hampel outlier filter, gap filling, aur train, validation aur test ka chronological split. DSM ₹ Calculator aur Transformer Health data-driven nahi hain; user value deta hai aur formula chalta hai, isliye unpar hamesha ILLUSTRATIVE likha hai. Synthetic data, jaise residential ya EV load profiles, sirf future scope pages ke liye hai; chhe live pages mein uska use nahi hai.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Har feature ka source, variables aur tag ek table mein.

Kyu kiya: Engineer ko pata hona chahiye ki kis number par kitna bharosa karein; real aur illustrative ko mix karna khatarnaak hai.

Kaise kiya: Field names seedhe app/schemas.py aur engine functions se liye; har chart par provenance badge.

Formula / number ka matlab

Row counts (DATA_QUALITY.md, data files)

5-minute grid par 293,184 slots chahiye the, 285,252 mile (2.705% missing). 15-minute par 97,728 blocks, jinme se 1,330 (1.361%) khaali chhode gaye, kyunki 3 ghante se lambe gap fill nahi karte. Usable blocks = 97,728 − 1,330 = 96,398.

Viva mein pooche ja sakte hain

Q: Causal cleaning kyun zaroori hai?

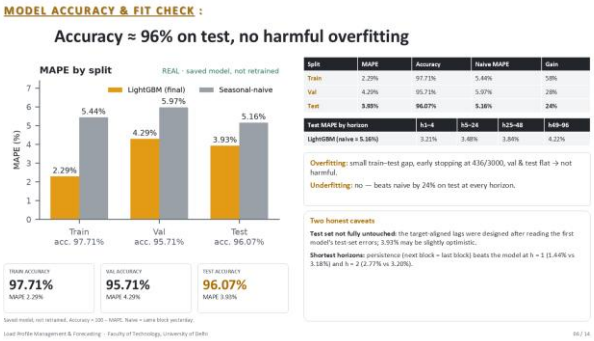
A: Wahi load series features aur targets dono banati hai. Agar cleaning future dekhe, to future ki jaankari features mein leak ho jaati hai aur score jhootha achha aata hai.

Q: Gap filling kaise kiya?

A: Gap ki lambai ke hisaab se: chhote gap forward-fill (894 blocks), medium gap pichhle 5 din ke same time ke profile se (1,153 blocks), aur 3 ghante se lambe gap khaali chhode.

Agli slide par: *Model kitna accurate hai, aur kya woh overfit ya underfit hai? Yeh agli slide batati hai.*

Slide 6 — Model Accuracy & Fit Check



Slide par kya hai

Left: train / validation / test MAPE ka chart (model vs seasonal-naive) aur teen accuracy tiles. Right: MAPE, accuracy, naive MAPE aur gain ki table; horizon-wise MAPE; overfitting / underfitting ka verdict; aur do honest caveats. Tag: REAL.

Bolne wali script (~75 sec)

Ab dekhte hain ki model kitna accurate hai, aur kya woh overfit ya underfit hai. Accuracy hum 100 minus MAPE se define karte hain. Test par MAPE 3.93% hai, yaani accuracy lagbhag 96%. Train par MAPE 2.29%, accuracy 97.71%; validation par 4.29%, accuracy 95.71%. Seasonal-naive, yaani kal isi block ka load, train par 5.44%, validation par 5.97% aur test par 5.16% deta hai, isliye model test par 24% kam galti karta hai. Overfitting check: train aur test ke beech gap hai, lekin chhota hai. Early stopping 3,000 mein se sirf 436 rounds par ruki, aur validation aur test error flat rahi; sirf train error giri. Isliye overfitting nuksaan wali nahi hai. Underfitting bhi nahi hai, kyunki model har horizon par naive se behtar hai: h = 1–4 par 3.21%, h = 5–24 par 3.48%, h = 25–48 par 3.84% aur h = 49–96 par 4.22%, jabki naive lagbhag 5.16%. Do honest caveats: pehla, test set poori tarah untouched nahi tha, kyunki target-aligned lags pehle model ki test errors dekh kar design hue, isliye 3.93% thoda optimistic ho sakta hai. Doosra, sabse chhote horizon par persistence model se behtar hai: h = 1 par 1.44% vs 3.18%.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Saved model ka train, validation aur test par accuracy aur overfitting / underfitting check.

Kyu kiya: Sirf ek accuracy number kaafi nahi; examiner poochhega ki model ratta to nahi maar gaya (overfit) ya bahut simple to nahi (underfit).

Kaise kiya: Saved model ko bina retrain kiye har split ke saare origin × horizon rows par score kiya; seasonal-naive se compare kiya; learning curve aur early-stopping rounds dekhe.

Formula / number ka matlab

Accuracy = 100 − MAPE; Gain = 1 − MAPE(model) / MAPE(naive)

Test: 100 − 3.93 = 96.07% ≈ 96%. Train 97.71%, validation 95.71%.

Gain: 1 − 3.93 / 5.16 = 24% (validation 28%, train 58%).

Early stopping: 436 rounds, limit 3,000 (patience 50).

Viva mein pooche ja sakte hain

Q: Overfitting hai ya nahi?

A: Train–test gap chhota hai (2.29% vs 3.93%), early stopping 436/3000 par ruki, aur validation aur test error flat hain, sirf train girti hai. Isliye overfitting nuksaan wali nahi hai.

Q: 96% accuracy kaise nikali?

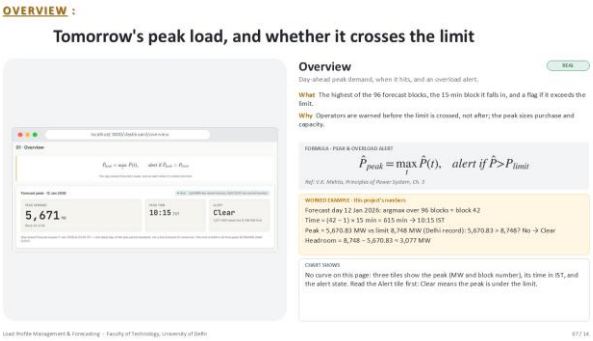
A: Accuracy = 100 − MAPE. Test MAPE 3.93% hai, to 100 − 3.93 = 96.07% ≈ 96%. Yeh classification accuracy nahi hai; har block ki galti ko actual load se divide karke average li gayi.

Q: Validation test se zyada kyun?

A: Validation ka do-tihai hissa March–June ki garmi hai, aur naive bhi wahan zyada galti karta hai (5.97% vs 5.16%); isliye woh window mushkil hai.

Agli slide par: *Ab ek-ek karke chhe live pages dekhte hain, shuru karte hain Overview se.*

Slide 7 — Overview



Slide par kya hai

Left: Overview page ka screenshot. Right: kya aur kyun, peak alert formula, worked example (12 Jan 2026), aur page kya dikhata hai. Tag: REAL.

Bolne wali script (~60 sec)

> Overview page sabse pehla sawal answer karta hai: kal ka peak kitna hoga, kab hoga, aur kya woh limit cross karega. Formula simple hai: 96 forecast blocks mein sabse bada value peak hai, aur agar woh limit se zyada hai to alert. Hamare test ke aakhri din, 12 January 2026 ka example: sabse bada forecast block 42 mein aaya. Block 42 ka start time hai 41 guna 15 minute, yaani 615 minute, matlab subah 10:15. Peak 5,670.83 MW hai. Limit humne Delhi ka record 8,748 MW rakhi. 5,670.83, 8,748 se kam hai, isliye alert 'Clear' hai, aur lagbhag 3,077 MW ka headroom bacha hai. Yeh forecast pichhli shaam 11:45 par issue hua tha, aur yeh backtest ka din hai, live kal ka forecast nahi.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Agle din ka peak, uska time, aur overload alert.

Kyu kiya: Operator ko limit cross hone se pehle warning milni chahiye, baad mein nahi.

Kaise kiya: LightGBM ke 96-block forecast par peak_and_alert engine chalta hai (ml/src/lpmf/engines/overview.py).

Formula / number ka matlab

P̂peak = max over t of P̂(t); alert agar P̂peak > Plimit

Step 1: 96 forecast values mein se sabse badi dhundho → block 42.

Step 2: time = (42 − 1) × 15 = 615 minute = 10:15 IST.

Step 3: 5,670.83 > 8,748? Nahi → alert = Clear.

Step 4: headroom = 8,748 − 5,670.83 ≈ 3,077 MW.

Viva mein pooche ja sakte hain

Q: Limit 8,748 MW hi kyun?

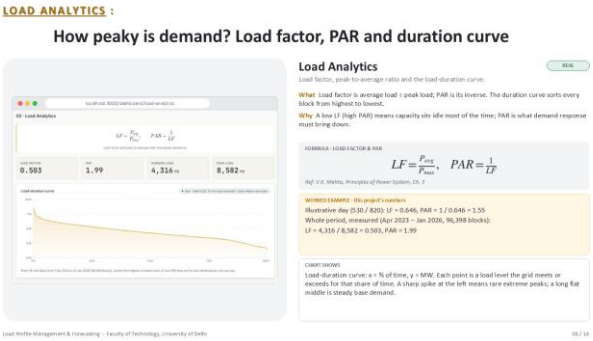
A: Yeh Delhi ka all-time record hai, isliye ek natural upper limit hai. Code mein limit ek parameter hai, DISCOM apni limit daal sakta hai.

Q: Do blocks ka value same ho to?

A: Engine sabse pehla (earliest) block chunta hai.

Agli slide par: *Peak ek number hai; poore din ki shape samajhne ke liye Load Analytics dekhte hain.*

Slide 8 — Load Analytics



Slide par kya hai

Load Analytics page: load factor, PAR aur load-duration curve. Worked example mein illustrative din aur poore period ke measured numbers dono. Tag: REAL.

Bolne wali script (~65 sec)

> Load factor, yaani LF, matlab average load ko peak load se divide karna. PAR, yaani peak-to-average ratio, iska ulta hai. LF kam ho to iska matlab hai ki grid ki capacity zyadatar time khaali padi rehti hai, sirf kuch ghante peak aata hai. Slide par do example hain, aur dono alag hain. Pehla illustrative din: average 530, peak 820, to LF = 530 / 820 = 0.646, aur PAR = 1.55. Yeh sirf formula samjhane ke liye hai. Doosra real, poore period ka measured: April 2023 se January 2026 ke 96,398 blocks. Average 4,316 MW, peak 8,582 MW, to LF = 0.503 aur PAR = 1.99. Graph load-duration curve hai: saare blocks ko bade se chhote load tak sort kiya. Left mein ek tez spike dikhti hai, matlab sabse zyada load bahut kam time ke liye aata hai.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Demand kitni peaky hai, LF, PAR aur duration curve se.

Kyu kiya: PAR wahi cheez hai jise demand response ko kam karna hota hai; isse capacity planning hoti hai.

Kaise kiya: load_factor_and_par engine; missing blocks skip hote hain, zero nahi maane jaate.

Formula / number ka matlab

LF = Pavg / Pmax; PAR = 1 / LF

Illustrative din: LF = 530 / 820 = 0.646 → PAR = 1 / 0.646 = 1.55.

Poora period (measured): LF = 4,316 / 8,582 = 0.503 → PAR = 1 / 0.503 = 1.99.

Duration curve: x-axis = kitne % time load is level ya usse upar raha, y-axis = MW.

Viva mein pooche ja sakte hain

Q: 0.646 aur 0.503 mein kaunsa sahi hai?

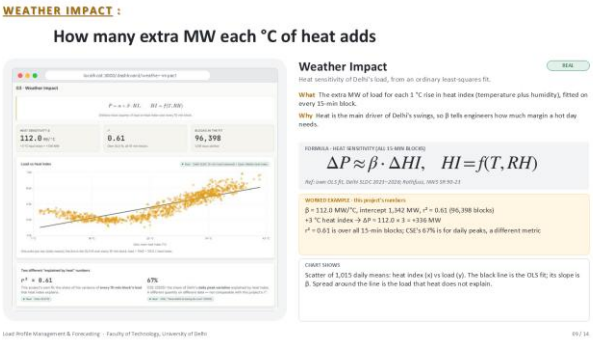
A: Dono sahi hain, lekin alag cheez hain. 0.646 ek illustrative din ka example hai; 0.503 poore data ka measured value hai. Dashboard 0.503 dikhata hai.

Q: Poore period ka LF ek din se kam kyun hai?

A: Lambe period mein peak saal ka sabse bada din hota hai, jabki average saare dino ka. Isliye ratio chhota aata hai.

Agli slide par: *Ab dekhte hain ki is peakiness ka sabse bada kaaran, garmi, load ko kitna badhata hai.*

Slide 9 — Weather Impact



Slide par kya hai

Weather Impact page: load vs heat index scatter, OLS line, β = 112.0 MW/°C, r^2 = 0.61. Tag: REAL.

Bolne wali script (~65 sec)

Yeh page batata hai ki har ek degree heat index badhne par load kitna badhta hai. Humne ordinary least squares, yaani OLS, se ek seedhi line fit ki: load = α + β × heat index. OLS matlab aisi line jo saare points se total squared doori sabse kam rakhe. Yeh fit 96,398 fifteen-minute blocks par hua. Result: β = 112.0 MW per degree Celsius. Iska matlab heat index 3 degree badhe to load lagbhag 112 × 3 = 336 MW badhta hai. r^2 = 0.61, matlab load ke variation ka 61% heat index se explain hota hai. Dhyaan dijiye, CSE ka 67% alag hai: woh daily peak par hai, hamara har 15-minute block par. Graph mein 1,015 dino ke daily average points hain aur kaali line OLS fit hai. Line ki dhalaan hi β hai.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Garmi se load kitna badhta hai, MW per °C mein.

Kyu kiya: Heat sabse bada driver hai; β se pata chalta hai ki garam din ke liye kitna margin chahiye.

Kaise kiya: heat_sensitivity engine har baar data se β dobara nikalta hai; heat index Rothfusz formula se, temp_c aur rh_pct se.

Formula / number ka matlab

$P = \alpha + \beta \cdot HI$; $\Delta P \approx \beta \cdot \Delta HI$; $HI = f(T, RH)$

β = 112.0 MW/°C, intercept α ≈ 1,342 MW, r^2 = 0.61.

+3 °C heat index → ΔP ≈ 112.0 × 3 = +336 MW.

Viva mein pooche ja sakte hain

Q: Graph mein curve U-shape jaisi kyun hai, line seedhi kyun?

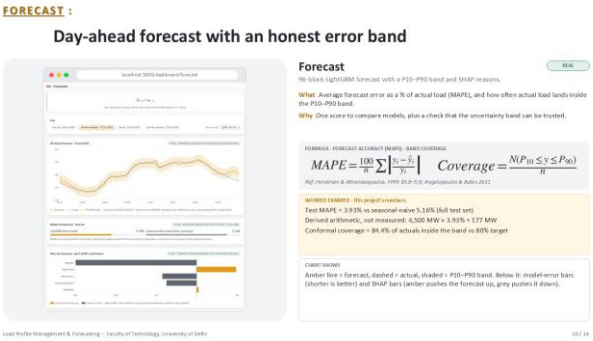
A: Thandi mein heating ki wajah se load thoda badhta hai, isliye neeche U jaisa dikhta hai. Humne ek simple linear fit rakha; isliye r^2 0.61 hai, 1 nahi. Non-linear fit future improvement ho sakta hai.

Q: Scatter daily means ka hai, fit blocks par kyun?

A: 96,398 points plot karne se graph bhar jaata, isliye 1,015 daily points dikhaye. β aur r^2 sab blocks par nikle hain, jaise slide par likha hai.

Agli slide par: *Ab project ka core: Forecast page.*

Slide 10 — Forecast



Slide par kya hai

Forecast page: 96-block forecast, P10–P90 band, actual load, model comparison aur SHAP. Formulas: MAPE aur band coverage. Tag: REAL.

Bolne wali script (~70 sec)

> Yeh project ka dil hai. LightGBM ek tree-based machine learning model hai jo bahut saare chhote decision trees ko jodkar prediction karta hai. Hum ek hi model se agle din ke saare 96 blocks predict karte hain. Accuracy MAPE se naapte hain, yaani average percentage error. Poore test set par, 13 July 2025 se 12 January 2026, MAPE 3.93% hai. Seasonal-naive baseline, yaani 'kal isi block ka load', ka MAPE 5.16% hai; error lagbhag 24% kam hua. Ek calculation se samjhiye: 4,500 MW ke block par 3.93% lagbhag 177 MW hota hai. Yeh sirf arithmetic hai, measured number nahi. Har forecast ke saath P10–P90 band hai, yaani ek 80% range. Target tha ki 80% actual values band ke andar aayein; test par 84.4% aayi. Graph mein amber line forecast hai, dashed line actual, aur shaded hissa band. Neeche model comparison aur SHAP bars hain jo batate hain ki forecast kyun aaya.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Agle din ke 96 blocks ka forecast, uncertainty band ke saath.

Kyu kiya: DSM charge forecast error par lagta hai; isliye accurate forecast aur imaandaar uncertainty dono chahiye.

Kaise kiya: Multi-horizon LightGBM (37 features, Optuna se tune, validation par early stopping), MAPIE split-conformal band, TreeSHAP.

Formula / number ka matlab

MAPE = (100/n) Σ |y − ŷ| / y; Coverage = N(P10 ≤ y ≤ P90) / n

MAPE: har block ki galti ko actual se divide karo, average lo, 100 se guna karo → 3.93%.

Derived (measured nahi): 4,500 MW × 3.93% ≈ 177 MW.

Coverage: band ke andar aaye actuals / total = 84.4% (target 80%).

Viva mein pooche ja sakte hain

Q: Error horizon ke saath kaise badalta hai?

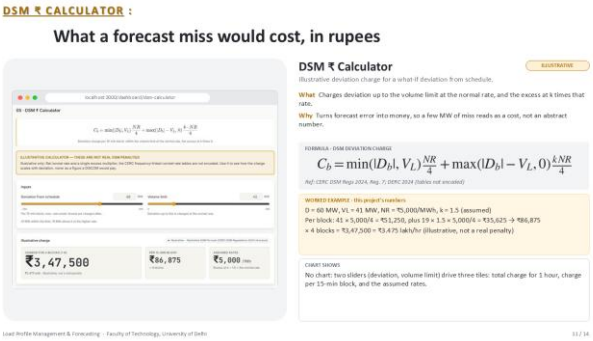
A: h = 1–4 par 3.21%, h = 5–24 par 3.48%, h = 25–48 par 3.84% aur h = 49–96 par 4.22%. Seasonal-naive har horizon par lagbhag 5.16% rehta hai, isliye model har horizon par behtar hai.

Q: Sabse important feature kaunsa nikla?

A: tlag_1day, yaani kal isi block ka load: 66.5% gain importance. Ise shuffle karne par MAPE 3.93% se 23.4% ho jaata hai, matlab yeh genuine signal hai, leak nahi.

Agli slide par: *Forecast mein galti hogi hi; agla page batata hai ki us galti ki keemat kya ho sakti hai.*

Slide 11 — DSM ₹ Calculator



Slide par kya hai

DSM ₹ Calculator page: do sliders (deviation, volume limit) aur teen result tiles. Formula aur ₹3.475 lakh/hr ka worked example. Tag: ILLUSTRATIVE.

Bolne wali script (~60 sec)

> DSM ka matlab Deviation Settlement Mechanism: agar DISCOM schedule se zyada ya kam bijli le, to charge lagta hai. Hamara formula: volume limit tak ka deviation normal rate NR par, aur limit se upar wala hissa k guna rate par. 4 se divide isliye karte hain kyunki 15 minute ek ghante ka chautha hissa hai, to MW ko MWh mein badalna hota hai. Example: deviation 60 MW, limit 41 MW, NR ₹5,000 per MWh, k = 1.5. Ek block mein: 41 × 5,000 / 4 = ₹51,250, aur bacha hua 19 MW × 1.5 × 5,000 / 4 = ₹35,625. Total ₹86,875 per block. Ek ghante mein 4 blocks, to ₹3,47,500, yaani ₹3.475 lakh per hour. Yeh illustrative hai, real penalty nahi, kyunki CERC ke frequency-linked rate tables hamare code mein encode nahi hain.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Forecast deviation ko rupees mein badalna.

Kyu kiya: Kuch MW ki galti abstract lagti hai; rupees mein dikhane par uski keemat samajh aati hai.

Kaise kiya: dsm_penalty engine; over- aur under-drawal dono |D| se charge; output hamesha is_illustrative = True.

Formula / number ka matlab

Cb = min(|D|, VL) · NR/4 + max(|D| − VL, 0) · k · NR/4

D = 60 MW, VL = 41 MW, NR = ₹5,000/MWh, k = 1.5 (assumed).

Limit tak: 41 × 5,000 / 4 = ₹51,250.

Limit se upar: (60 − 41) × 1.5 × 5,000 / 4 = 19 × 1,875 = ₹35,625.

Per block: ₹86,875 → 4 blocks (1 ghanta): ₹3,47,500 = ₹3.475 lakh/hr.

Viva mein pooche ja sakte hain

Q: Yeh number real penalty kyun nahi?

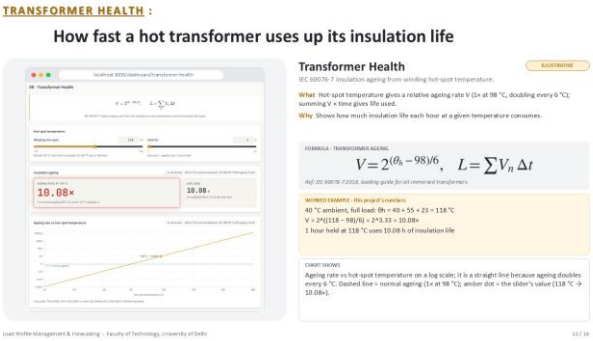
A: CERC 2024 mein normal rate grid frequency se judi hoti hai; woh tables encode nahi kiye. Humne ek flat rate aur ek multiplier maana hai, isliye sirf charge ki shape dikhti hai.

Q: 41 MW limit kahan se?

A: Yeh BRPL ke liye DERC 2024 intra-state DSM order wali volume limit ke roop mein li gayi hai, aur NR = ₹5,000 assumed hai. Calculator mein user limit ko slider se badal sakta hai.

Agli slide par: *Paise ke baad equipment: garmi mein transformer kitna jaldi budha hota hai.*

Slide 12 — Transformer Health



Slide par kya hai

Transformer Health page: hot-spot temperature slider, ageing rate 10.08×, aur log-scale ageing curve. IEC 60076-7 formula. Tag: ILLUSTRATIVE.

Bolne wali script (~60 sec)

Transformer ki insulation garmi se budhi hoti hai. IEC 60076-7 standard ke hisaab se 98 degree hot-spot par ageing rate 1 hai, yaani normal. Har 6 degree badhne par yeh double ho jaata hai. Hot-spot matlab winding ka sabse garam point. Example: 40 degree ambient aur full load par hot-spot = 40 + 55 + 23 = 118 degree. Tab V = 2 ki power (118 − 98) / 6, yaani 2 ki power 3.33, jo 10.08 hai. Matlab 118 degree par ek ghanta chalane se insulation ki 10.08 ghante ki life khatam hoti hai. Graph log scale par hai, isliye seedhi line dikhti hai. Dashed line normal ageing hai, aur amber dot slider ki value. Yeh page what-if hai, kisi real transformer ka measured data nahi, isliye ILLUSTRATIVE.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Hot-spot temperature se insulation ageing aur life used.

Kyu kiya: Garam dino mein kaunsa ghanta transformer ko sabse zyada nuksaan karta hai, yeh dikhana.

Kaise kiya: transformer_ageing engine; 118 °C default steady_state_hotspot_c se (ΔθOr = 55 K, H·gr = 23 K, K = 1).

Formula / number ka matlab

V = 2^((θh − 98)/6); L = Σ Vn · Δt

θh = 40 + 55 + 23 = 118 °C (40 °C ambient, rated load).

V = 2^((118 − 98)/6) = 2^3.33 = 10.08×.

1 ghanta × 10.08 = 10.08 ghante ki life use.

Viva mein pooche ja sakte hain

Q: 55 aur 23 kahan se aaye?

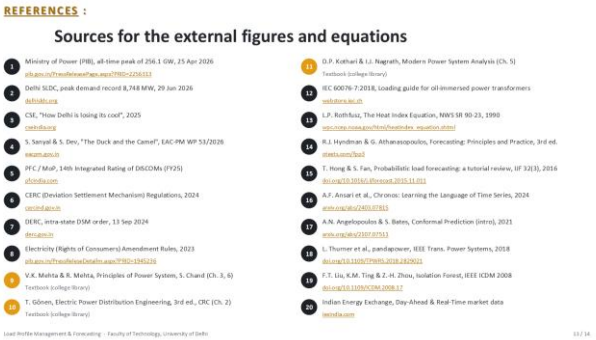
A: IEC 60076-7 ke distribution transformer (ONAN) defaults: top-oil rise 55 K aur hot-spot gradient 23 K. Rated load par yeh ambient se 78 K upar le jaate hain.

Q: Yeh page real kyun nahi?

A: Hamare paas kisi transformer ka temperature data nahi hai; user temperature deta hai aur formula chalta hai.

Agli slide par: *Yeh the hamare chhe live pages. Ab un sources ko dekhte hain jinse yeh numbers aur formulas aaye.*

Slide 13 — References


REFERENCES :

Sources for the external figures and equations

1 Ministry of Power (PIB), all-time peak of 256.1 GW, 25 Apr 2026
pib.gov.in/PressReleasePage.aspx?PRID=2256333

2 Delhi SLDC, peak demand record 8,748 MW, 29 Jun 2026
delhisldc.org

3 CSE, "How Delhi is losing its cool", 2025
cseindia.org

4 S. Sanyal & S. Dev, "The Duck and the Camel", EAC-PM WP 53/2026
eac.gov.in

5 PFC / MoP, 14th Integrated Rating of DISCOMs (FY25)
pfcindia.com

6 CERC (Deviation Settlement Mechanism) Regulations, 2024
cercind.gov.in

7 DERC, intra-state DSM order, 13 Sep 2024
derc.gov.in

8 Electricity (Rights of Consumers) Amendment Rules, 2023
pib.gov.in/PressReleaseDetailm.aspx?PRID=1945216

9 V.K. Mehta & R. Mehta, Principles of Power System, S. Chand (Ch. 3, 6)
Textbook (college library)

10 T. Gönen, Electric Power Distribution Engineering, 3rd ed., CRC (Ch. 2)
Textbook (college library)

11 D.P. Kothari & I.J. Nagrath, Modern Power System Analysis (Ch. 5)
Textbook (college library)

12 IEC 60076-7:2018, Loading guide for oil-immersed power transformers
webstore.iec.ch

13 L.P. Rothfusz, The Heat Index Equation, NWS SR 90-23, 1990
wpc.ncep.noaa.gov/html/heatindex_equation.shtml

14 R.J. Hyndman & G. Athanasopoulos, Forecasting: Principles and Practice, 3rd ed.
otexts.com/fpp3

15 T. Hong & S. Fan, Probabilistic load forecasting: a tutorial review, IJF 32(3), 2016
doi.org/10.1016/j.ijforecast.2015.11.011

16 A.F. Ansari et al., Chronos: Learning the Language of Time Series, 2024
arxiv.org/abs/2403.07815

17 A.N. Angelopoulos & S. Bates, Conformal Prediction (intro), 2021
arxiv.org/abs/2107.07511

18 L. Thurner et al., pandapower, IEEE Trans. Power Systems, 2018
doi.org/10.1109/TPWRS.2018.2829021

19 F.T. Liu, K.M. Ting & Z.-H. Zhou, Isolation Forest, IEEE ICDM 2008
doi.org/10.1109/ICDM.2008.17

20 Indian Energy Exchange, Day-Ahead & Real-Time market data
iexindia.com

Load Profile Management & Forecasting — Faculty of Technology, University of Delhi 13 / 14

Slide par kya hai

20 sources: PIB, Delhi SLDC, CSE, EAC-PM, CERC, DERC, IEC 60076-7, textbooks (V.K. Mehta, Kothari & Nagrath, Gönen), Hyndman & Athanasopoulos (FPP3), conformal prediction, Chronos, IEX.

Bolne wali script (~25 sec)

> Slide par har external number aur equation ka source hai. Data ke liye Delhi SLDC aur Open-Meteo; regulation ke liye CERC DSM 2024 aur DERC; transformer ke liye IEC 60076-7; formulas ke liye V.K. Mehta aur Kothari & Nagrath; aur forecasting ke liye Hyndman ki FPP3 aur conformal prediction ka paper.

Kya hua / Kyu kiya / Kaise kiya

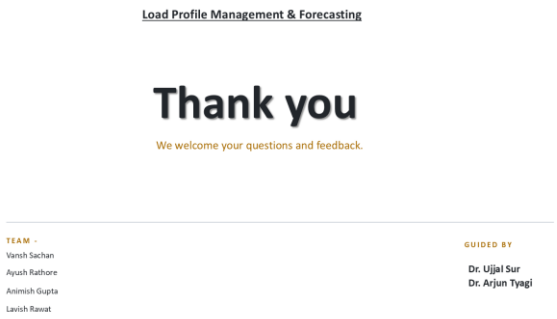
Kya hua: Saare sources ek jagah.

Kyu kiya: Har number verify kiya ja sake.

Kaise kiya: Link aur textbook chapter ke saath.

Agli slide par: *Aur ab, closing.*

Slide 14 — Thank You



Slide par kya hai

Thank you, team aur guides ke naam.

Bolne wali script (~25 sec)

> Toh sir/ma'am, humne ek real Delhi forecast ko uske nateejon tak le jaane ki koshish ki: peak alert, demand ki shape, garmi ka asar, DSM charge aur transformer ageing, aur har number par saaf label. Aapke time ke liye dhanyavaad. Ab hum aapke sawaalon ka swagat karte hain.

Kya hua / Kyu kiya / Kaise kiya

Kya hua: Closing.

Kyu kiya: Ek line mein project yaad dilana.

Kaise kiya: Sawaalon ke liye invite.

Viva ke tough sawal

Har jawab ka number deck, report ya repo (MODEL_RESULTS.md, DATA_QUALITY.md, engine code) se match karta hai.

1. Naive baseline kya hai aur kyun zaroori hai?

Seasonal-naive ka matlab: kal isi 15-minute block ka load hi aaj ka forecast. Isme koi learning nahi hai. Agar hamara model isse behtar nahi, to model ka koi fayda nahi. Test par naive 5.16% aur LightGBM 3.93% hai, aur model har horizon par naive se behtar hai.

2. LightGBM hi kyun? SARIMAX kyun drop hua?

LightGBM lags, weather aur calendar jaise bahut saare features ek saath le sakta hai, fast hai, aur SHAP se explain hota hai. SARIMAX (hourly) code mein bana hai, lekin validation par order search mein sabse achha result bhi lagbhag 25.3% MAPE tha, aur uska test par score kabhi nahi hua. Isliye woh reported results mein nahi hai.

3. Leakage kya hota hai? Hampel wala bug kaise pakda aur fix kiya?

Leakage matlab model ko training mein aisi jaankari mil jaana jo forecast ke time par hoti hi nahi, jaise future ka data. Pehle Hampel outlier filter ek centred window use karta tha: kisi reading ko outlier kehne ke liye 2 ghante aage tak ki readings dekhta tha, aur sigma par koi floor nahi tha, isliye 11.5% real readings flag ho rahi thi. Ek independent code review mein yeh pakda gaya. Fix: trailing (sirf peeche dekhne wali) 35-minute window, aur sigma par median ke 2% ka floor. Ab sirf 0.1% readings flag hoti hain. Profile-fill ka ek aur fix ke saath MAPE 3.92% hua, jo chhota (~0.07 pp) aur uncertain step tha.

4. Naive ke 5.22% aur 5.16% mein farak kya hai?

Data same hai, rows alag hain. 5.16% poore test set par hai: woh 1,656,240 rows jinke liye kal ka value maujood hai. 5.22% model ladder table mein hai, jo sirf 1,267,749 'common real' rows par hai, yaani woh rows jo har data version mein real reading thi. Isliye dono numbers apni-apni table mein label ke saath hain.

5. LF 0.646 aur 0.503 mein farak?

0.646 ek illustrative din hai: 530 / 820, sirf formula samjhane ke liye. 0.503 poore period (April 2023 – January 2026, 96,398 blocks) ka measured value hai: 4,316 / 8,582. Dashboard 0.503 dikhata hai.

6. P10–P90 ka target 80% hai aur coverage 84.4% hai. Iska matlab?

P10–P90 ek 80% range hai: theory mein 80% actual values iske andar aani chahiye. Test par 84.4% aayi, yaani band thoda conservative (chauda) hai, kyunki validation period (March–June heavy) ki errors test se badi thi. Yeh average hai: August mein 73% aur November mein 98%.

7. Test set fully untouched kyun nahi raha?

Pehle LightGBM ki test-set errors dekh kar (woh lambe horizons par naive se haar raha tha) humne target-aligned lag features design kiye. Hyperparameters aur early stopping ne test kabhi nahi dekha, lekin feature design par test ka asar pada. Isliye 3.93% thoda optimistic ho sakta hai; hum yeh khud batate hain.

8. h = 1 par persistence model se behtar kyun hai?

15 minute mein load bahut kam badalta hai (lag-1 block autocorrelation 0.998). Isliye 'agla block = abhi wala block' h = 1 par 1.44% MAPE deta hai, jabki model 3.18%. Hamara ek hi model saare 96 horizons ke liye optimise hua hai. h = 2 par bhi persistence jeetta hai (2.77% vs 3.20%). Fix: chhote horizons par persistence ya alag short-horizon model.

9. Diwali par error zyada kyun hai?

Diwali (20 October 2025) par day MAPE 8.65% tha aur sirf 49% blocks band mein aaye. Actual load 4-week average (5,530 MW) se lagbhag 2,000 MW kam tha. Model holiday ko pehchanta hai, lekin training mein holidays bahut kam the aur sirf ek indicator feature hai, isliye effect bahut chhota seekha.

10. Synthetic aur illustrative data kahan use hua aur kyun?

Synthetic: paanch consumer category profiles (residential, commercial, industrial, agricultural, EV), parametric model se bane, sirf future-scope pages ke liye. Illustrative: DSM ₹ Calculator, Transformer Health aur LF ka 530/820 example. Chhe live pages ka baaki sab real data hai.

11. DSM ₹ figure illustrative kyun hai?

CERC DSM Regulations 2024 mein normal rate grid frequency par depend karta hai. Woh tables encode nahi kiye; humne flat ₹5,000/MWh aur k = 1.5 maana. Isliye ₹3.475 lakh/hr sirf charge ki shape dikhata hai, real bill nahi.

12. 8,748 MW wala peak dataset mein kyun nahi?

Dataset 12 January 2026 par khatam hota hai, aur yeh record 29 June 2026 ka hai. Data ka maximum 8,581.6 MW (19 June 2024) hai. Isliye test window mein March–June ki garmi bhi nahi hai, aur summer peak accuracy test nahi hui.

13. Overfitting hai ya nahi?

Train, validation aur test MAPE 2.29%, 4.29% aur 3.93% hain (accuracy 97.71%, 95.71%, 96.07%). Gap chhota hai: early stopping ne 3,000 mein se 436 rounds rakhe, aur validation aur test error flat rahi, sirf train error giri. Validation zyada mushkil hai kyunki uska do-tihai hissa March–June hai; naive mein bhi same gap hai (5.97% vs 5.16%). Underfitting bhi nahi: model test par naive se 24% behtar hai, har horizon par.

14. Data clean karne mein kitna badla gaya?

15-minute ke 97,728 blocks mein se 2,047 fill kiye gaye (894 forward-fill, 1,153 profile-fill), 249 outlier flag hue, aur 1,330 blocks khaali chhode. Har filled block par flag hai, kuch bhi chhupa nahi.

15. 96% accuracy kaise nikali?

Accuracy = 100 − MAPE. Test MAPE 3.93% hai, isliye accuracy 96.07% (≈ 96%); train par 97.71%, validation par 95.71%. Yeh classification accuracy nahi hai: har 15-minute block ki galti ko actual load se divide karke average li, phir 100 se ghataya. Seasonal-naive ki accuracy 94.84% (MAPE 5.16%) hai, isliye asli fayda 24% kam galti hai.